

Do Connected Neurons Share Function?

Pair-level and topology analysis of the MICrONS dataset

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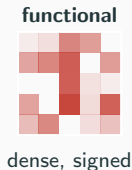
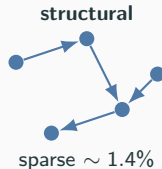
Neuroscience Project

The question

Do structurally connected neurons share function?

Two networks over the same $N=906$ cortical neurons.

- **Structural graph C**: EM-reconstructed, directed, sparse, synapse-weighted.
- **Functional graph F^{corr}** : two-photon signal correlation, dense, symmetric, signed.



Central question

Do directly connected pairs share function, and do the two networks have similar *topology*?

Builds on Ko et al. (2011), Cossell et al. (2015), Ding et al. (2025), and the MICrONS Consortium (2025), which established like-to-like wiring in mouse visual cortex.

Setup

Cohort and data

- **MICrONS** EM volume + co-recorded two-photon imaging.
- $N = 906$ functionally matched, proofread excitatory neurons.
- $M = 116$ shared oracle response conditions.
- All 906 have valid functional responses.

Convention

$C[i, j] > 0 \Leftrightarrow$ directed synaptic edge from i (pre) to j (post).

Pair-level tests use unordered pairs $i < j$.

Area	Neurons	Within-area conn. pairs
V1	728	10,441
RL	122	312
AL	56	91
cross	n/a	736

Structural graph C

directed edges	11,822
directed density	0.0144

Pair table

unordered pairs	409,965
connected	11,580
none / uni / bi	398,385 / 11,338 / 242

Defining functional and structural similarity

Functional similarity: Pearson correlation on shared responses $F \in \mathbb{R}^{N \times M}$:

$$F_{ij}^{\text{corr}} = \frac{\sum_c (F_{ic} - \bar{F}_i)(F_{jc} - \bar{F}_j)}{\sqrt{\sum_c (F_{ic} - \bar{F}_i)^2 \sum_c (F_{jc} - \bar{F}_j)^2}}.$$

Structural similarity: binary adjacency $\mathbf{A}_{ij} = \mathbf{1}[\mathbf{C}_{ij} \vee \mathbf{C}_{ji}]$ and weighted strength $\sum(\text{synapse size})$.

Pair non-independence

Every neuron appears in $N-1$ pair rows. Mann–Whitney p -values are not confirmatory; the headline test is a **neuron-identity permutation null**: hold $\mathbf{A}$ and F^{corr} fixed, shuffle neuron labels, recompute the contrast under 10,000 permutations.

1. **Pair level.**

Are directly connected neurons more functionally correlated than unconnected neurons?

Tested by H1; refined by H2–H6.

2. **Network level.**

Do the structural and functional networks share broader topology?

Tested by H7 and thresholded-graph comparisons.

Hypotheses

- H1.** Are connected pairs more correlated?
- H2.** Synapse-strength gradient
- H3.** Reciprocity
- H4.** Is distance the real driver?
- H5.** Composition controls (area, layer, cell type)
- H6.** Orientation-tuning similarity
- H7.** Are structural hubs functional hubs?

Pair-level results

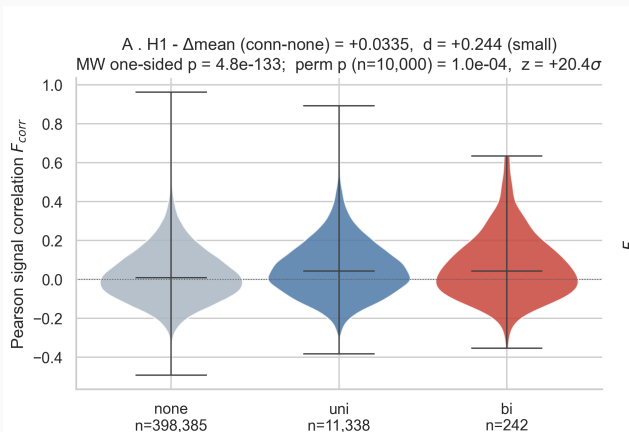
H1. Are connected pairs more correlated?

- **Pair classes:** unconnected, unidirectional, bidirectional.
- **Metric:** pairwise signal correlation F^{corr} .
- **Test:** mean-difference with bootstrap 95% CI and one-sided Mann–Whitney.
- **Null:** neuron-identity permutation that shuffles labels but holds the structural graph fixed.

Why the permutation null?

Pair rows are not independent; every neuron appears in many pairs. Conventional p -values understate uncertainty; the permutation null respects this.

H1. Connected pairs are more correlated



$\overline{F_{\text{corr}}}$ by class:

- none = $+0.0223$ ($n=398,385$)
- uni = $+0.0554$ ($n=11,338$)
- bi = $+0.0680$ ($n=242$)

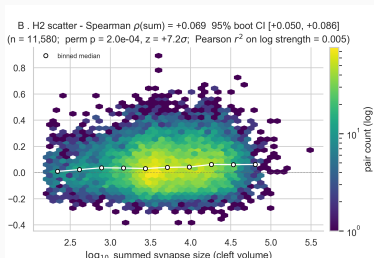
$\Delta\overline{F_{\text{corr}}} \text{ (conn-none)} = +0.0335$,
95% CI [$+0.031$, $+0.036$],
Cohen's $d = +0.244$ (small).

Permutation null ($n_{\text{perm}} = 10,000$):
observed at $z = +20.4\sigma$, $p \leq 10^{-4}$.

Take-away

Anatomy shifts the functional similarity distribution *on average*; it does not determine any single pair.

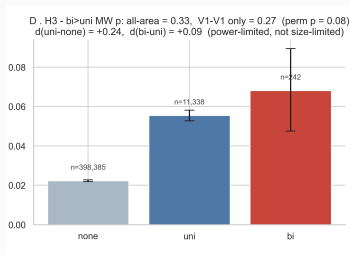
H2. Synapse-strength gradient



Spearman $\rho = +0.069$, CI [+0.050, +0.086].

Strength explains < 1% of F^{corr} variance.

H3. Reciprocity



uni > none: $d = +0.24$.

bi > uni: $d = +0.09$, not resolved.

Same null in V1-V1-only retest.

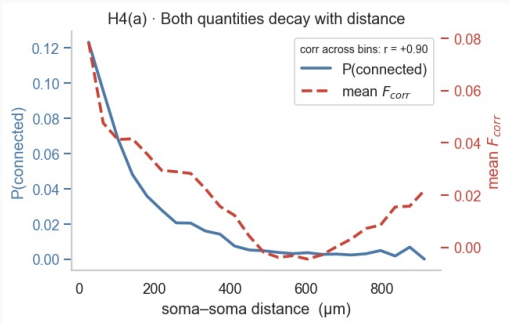
Take-away

Edge weight nudges F^{corr} upward but explains little variance; bidirectionality adds no detectable premium beyond being connected.

Three worries for the H1 effect

H1 says connected pairs are more correlated.
Three obvious confounds could explain that
without a real structure–function link:

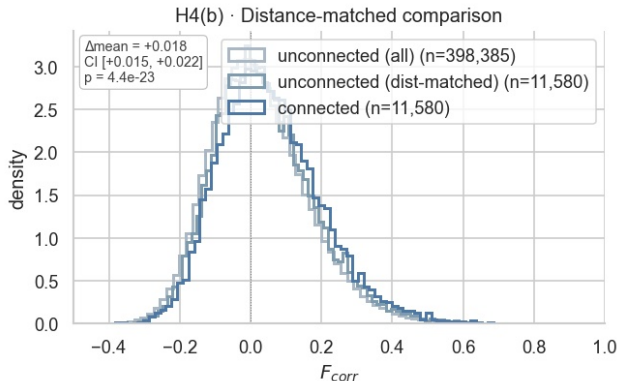
- **Distance (H4).** Connection probability and signal correlation *both* decay with soma–soma distance; connected pairs sit at short range, where F^{corr} is naturally higher.
- **Composition (H5).** Pairs sharing area, layer, or cell type are both more often connected and more often co-tuned.
- **Orientation tuning (H6).** Like-to-like wiring in V1 is classically described in terms of orientation preference — is F^{corr} just a noisy proxy?



Across 24 distance bins, $P(\text{connected})$ and mean F^{corr} co-decay at $r = +0.90$.

The shared geometry is real — H4 controls for it first.

H4. Distance matching: connected vs distance-matched controls



For each of 11,580 connected pairs, take the closest-distance unconnected pair as a control (no replacement).

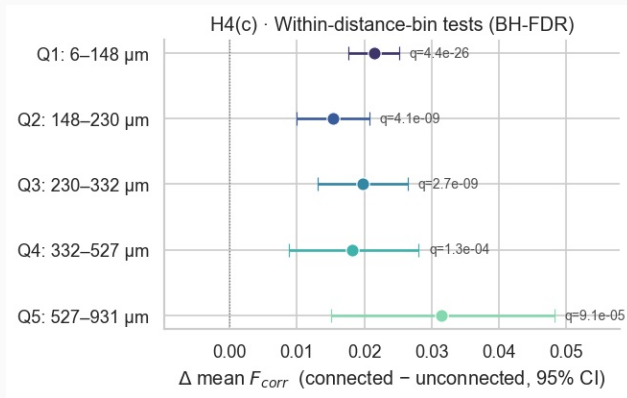
$$\Delta \overline{F^{corr}}_{matched} = +0.018, \\ 95\% \text{ CI } [+0.015, +0.022], p = 4.4 \times 10^{-23}.$$

Non-parametric: no functional form assumed for the F^{corr} -distance relationship.

Take-away

Matched against distance alone, the connectivity premium remains clearly positive.

H4. Within-distance-bin tests (BH-FDR corrected)



Split pairs into five distance quintiles; retest H1 inside each.

BH-FDR controls expected false-discovery rate across the five tests.

All five bins positive; every one significant after BH-FDR ($q \leq 10^{-4}$ throughout).

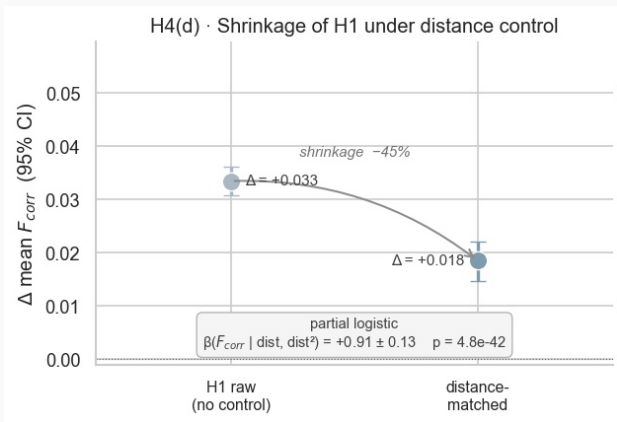
Within-bin $\Delta \overline{F}^{corr}$ ranges from +0.014 to +0.031.

Effect persists from $\sim 6 \mu\text{m}$ up to $\sim 931 \mu\text{m}$.

Take-away

The connectivity premium holds locally at every distance — not driven by a single regime.

H4. Distance shrinks but does not erase the effect



Distance-matched contrast.

$$\Delta \overline{F^{corr}} : \underbrace{+0.033}_{\text{raw}} \rightarrow \underbrace{+0.019}_{\text{matched}}$$

~ 45% shrinkage; effect remains highly significant.

All five within-distance-bin retests positive.

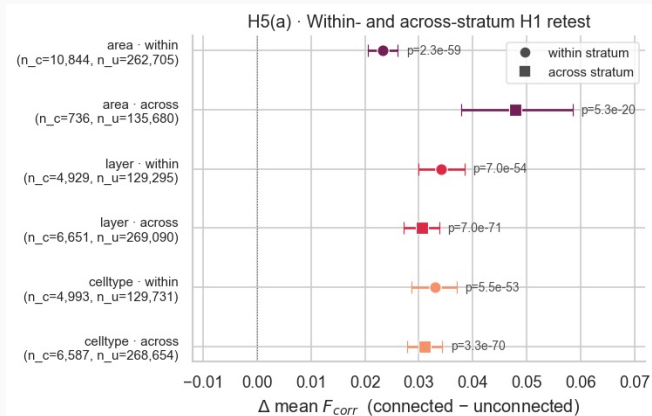
Partial logistic

$$\beta(F^{corr} | \text{dist}, \text{dist}^2) = +0.91 \pm 0.13, \\ p = 4.8 \times 10^{-42}.$$

Take-away

Distance explains part of the gap; the connected-pair effect survives.

H5. Stratified retests across area, layer, cell type



Retest H1 separately within and across each compositional split (six contrasts in total).

All six contrasts positive; every one significant ($p < 10^{-19}$ throughout).

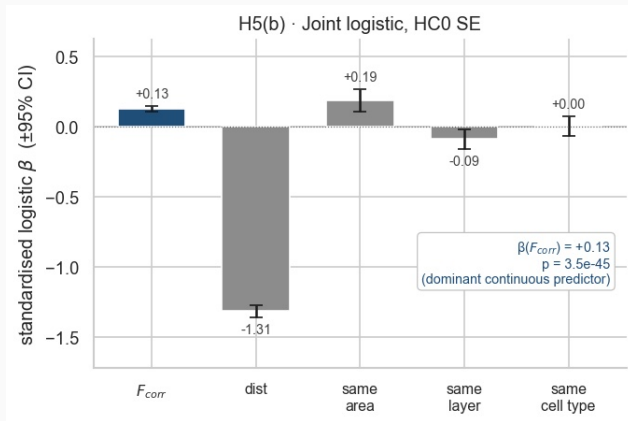
Within-stratum Δ ranges +0.023 to +0.034.

Across-stratum Δ up to +0.048.

Take-away

The effect is not driven by any single area, layer, or cell-type pool.

H5. Joint logistic with composition controls



Logistic on connectivity with five standardised predictors:

$z(F^{corr})$, $z(dist)$, same-area, same-layer, same-celltype.

$\beta(F^{corr}) = +0.13$, $p = 3.5 \times 10^{-45}$:
dominant continuous positive predictor.

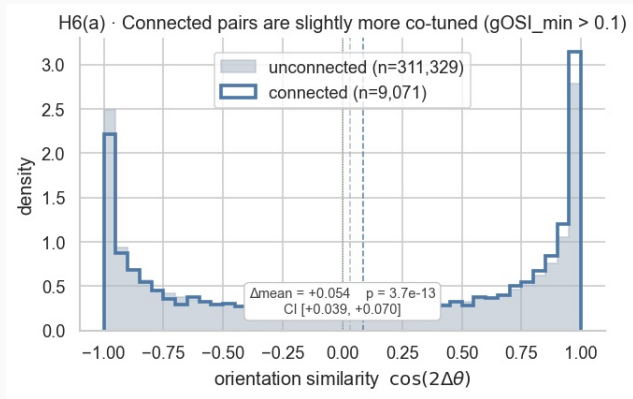
Distance dominates magnitude
($\beta = -1.31$), as expected.

same-area +0.19; same-layer -0.09;
same-celltype ≈ 0 .

Take-away

F^{corr} survives composition controls — not a within-compartment artifact.

H6. Connected pairs are slightly more co-tuned



Restrict to well-tuned pairs (gOSI > 0.10 for both cells; 320,400 pairs).

Orientation similarity:

$$\text{ori_sim} = \cos(2(\theta_i - \theta_j)) \in [-1, +1].$$

$$\overline{\Delta\text{ori_sim}}_{\text{conn-none}} = +0.054,$$

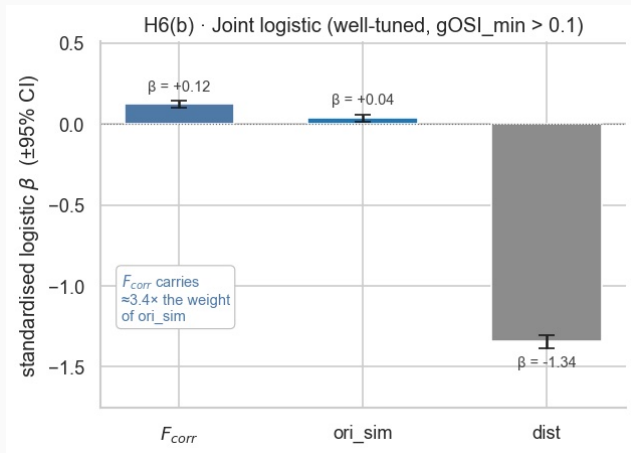
$$95\% \text{ CI } [+0.039, +0.070],$$

$$p = 3.7 \times 10^{-13}.$$

Take-away

Connected pairs really are more co-tuned —
so is F^{corr} just an orientation proxy?

H6. F^{corr} beats orientation similarity



Joint logistic on the well-tuned subset:

$$z(F^{\text{corr}}) + z(\text{ori_sim}) + z(\text{dist}).$$

$$\beta(F^{\text{corr}}) = +0.12 \text{ vs } \beta(\text{ori_sim}) = +0.04.$$

$$\text{Ratio } \beta(F^{\text{corr}})/\beta(\text{ori_sim}) \approx 3.4.$$

Both positive, but F^{corr} carries the bulk of the predictive weight beyond orientation alone.

Take-away

F^{corr} is not orientation tuning in disguise — it captures broader shared response structure.

Network topology

From pairs to topology

The assignment also asks whether the two networks share *broader structure*.

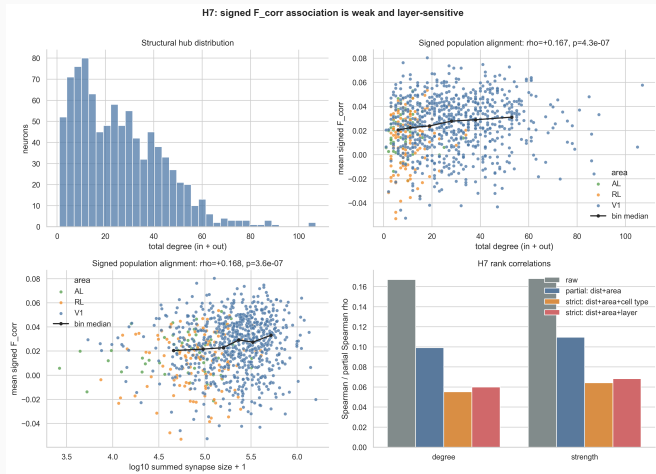
- **Node level:** degree, strength, mean functional coupling.
- **Graph level:** clustering, thresholded density, degree distribution.
- **Conceptual mismatch:** sparse directed synapses vs. dense symmetric correlations.

Two strategies

- Compare continuous node-level signals (H7, next).
- Threshold the functional graph at structural density and compare topology metrics.

H7. Are structural hubs functional hubs?

Reminder: sparse directed synapses vs dense symmetric correlations; comparison is non-trivial.



Primary (total_strength, dist+area controlled):

$\rho_{\text{partial}} = +0.110$, $\approx 1.2\%$ rank variance.

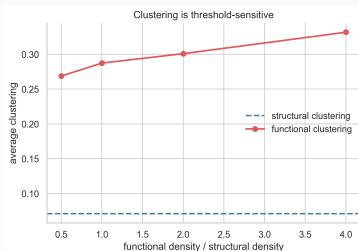
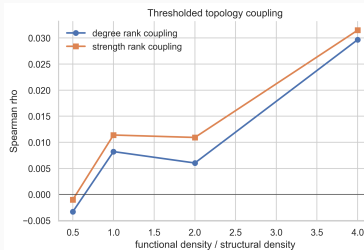
Adding cell type or layer: ρ drops to $\sim +0.06$, non-significant after BH-FDR.

Sign-agnostic $|F^{\text{corr}}|$: rank correlations near zero.

Take-away

A narrow signed-alignment effect, *not* a broad hub-to-hub correspondence.

Thresholded topology



Threshold the dense functional matrix at densities $\{0.5, 1, 2, 4\} \times$ structural density (0.028 undirected).

Degree- and strength-rank coupling stay *near zero* across thresholds (max $\rho \approx 0.03$, all $p > 0.3$).

Clustering coefficients differ by a factor of $\sim 4-5$: structural ≈ 0.071 vs functional 0.27–0.33.

Take-away

Sparse directed synapses and dense symmetric correlations are not topologically equivalent in this cohort.

Cross-check and scope

Cross-check: 93-neuron same-scan cohort

- Same animal, same recording session: V1 layer 4, scan 9.3.
- Smaller cohort but tighter measurement regime: trial-level repeats available.
- Three correlation measures isolate different aspects of co-firing:
 - **Trace:** raw per-trial responses (signal + noise).
 - **Signal:** per-stimulus means (tuning component).
 - **Noise:** residuals after subtracting stimulus means (shared variability).

H1 replicates in the same-scan cohort

$N = 93$ V1 L4 neurons, scan 9.3.

Trace correlation, 4,278 pairs.

$$\Delta \overline{F^{\text{corr}}}_{\text{conn-none}} = +0.026,$$

95% CI [+0.018, +0.034], $d = +0.71$.

Trace correlation has a different absolute scale

than the 906-cohort signal correlation; d values are not directly comparable.

Take-away

H1 holds in a different cohort and measurement regime; the effect persists in residual noise correlation.

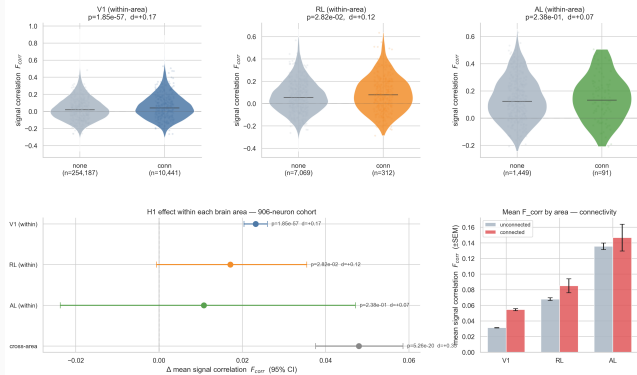
Signal/noise decomposition (n=93)

Measure	$\Delta \overline{F^{\text{corr}}}$	d
Trace	+0.026	+0.71
Signal	+0.053	+0.41
Noise	+0.047	+0.38

Effect appears in residual noise correlation $\Rightarrow$ not purely shared tuning.

V1/RL/AL area scope (n=906)

Figure 6 — V1 / RL / AL area comparisons (906-neuron signal-correlation cohort)



Area scope (n=906)

Area	$\Delta \overline{F_{corr}}$	d	
V1 within	+0.023	+0.17	robust
RL within	+0.017	+0.12	marginal
AL within	+0.011	+0.07	null
cross-area	+0.048	+0.35	

RL/AL underpowered (122 / 56 neurons).

Take-away
 Effect clearest in V1; cross-area pairs show the largest gap.

Discussion

Pair-level

Direct synaptic connectivity is associated with higher signal correlation. The effect is modest in absolute size but stays positive after distance, area, layer, cell type, and orientation tuning; it also appears in residual noise correlation in a same-scan cohort, so it is not only shared tuning.

Network-level

Structural and functional networks are *not* topologically equivalent in this cohort. Structurally central neurons do not generically become functionally central, and thresholded graph comparisons collapse to chance.

Consistent with Cossell et al. (2015), Ko et al. (2011), Ding et al. (2025), and the MICrONS Consortium (2025): like-to-like wiring carries pair-level functional information; topology equivalence is not the right framing for sparse-directed vs dense-symmetric networks.

Limitations

- **Pair non-independence:** mitigated by permutation null, not eliminated.
- **Cross-scan vs same-scan correlations:** not directly comparable.
- **Coarse composition labels:** transcriptomic substructure unmeasured.
- **Topology comparisons are threshold-dependent:** we report sensitivity, not a single graph.
- **Area imbalance:** V1 dominates; RL/AL claims are exploratory.

Connectivity predicts function pair-by-pair,
but the two networks are not topologically equivalent.

- **Cossell L., Iacaruso M. F., Muir D. R. et al.** (2015). Functional organization of excitatory synaptic strength in primary visual cortex. *Nature*, 518, 399–403.
- **Ko H., Hofer S. B., Pichler B. et al.** (2011). Functional specificity of local synaptic connections in neocortical networks. *Nature*, 473, 87–91.
- **Ding Z. et al.** (2025). Functional connectomics reveals general wiring rule in mouse visual cortex. *Nature*, 640, 459–469.
- **MICrONS Consortium** (2025). Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640, 435–447.

Data: MICrONS minnie65 volume + co-recorded two-photon imaging.

Backup: Results summary

Hypothesis	Effect	Verdict
H1. connected > unconnected	$\Delta \overline{F^{\text{corr}}} = +0.033$, $d = +0.24$, perm $z = +20.4\sigma$	supported
H2. synapse-strength gradient	Spearman $\rho = +0.069$	weak positive
H3. reciprocity premium	bi vs uni: $d = +0.09$, unresolved	partial
H4. distance control	matched $\Delta = +0.019$; all bins positive	supported
H5. composition control	joint $\beta(F^{\text{corr}}) = +0.13$	supported
H6. orientation tuning	$\beta(F^{\text{corr}}) \approx 3.4 \times \beta(\text{ori_sim})$	F^{corr} dominant
H7. hub coupling	narrow signed $\rho = +0.11$; $ F^{\text{corr}} $ null	narrow only
Thresholded topology	degree/strength rank coupling ~ 0	not detected
Cross-check, trace (n=93)	$\Delta = +0.026$, $d = +0.71$	consistent
Cross-check, signal (n=93)	$\Delta = +0.053$, $d = +0.41$	consistent
Cross-check, noise (n=93)	$\Delta = +0.047$, $d = +0.38$	beyond tuning
Area scope V1 / RL / AL	V1 robust; RL marginal; AL underpowered	V1 clearest